$v = 5.886 = 5.9 \text{ m s}^{-1}$

5 **A** Since the resultant force on the ball is zero, it falls with constant velocity. Thus, its kinetic energy remains constant with time.

Its gravitational potential energy will decrease linearly with time as its height is decreasing at a constant rate due to its constant downward velocity.

$$\frac{E_p}{t} = \frac{mgh}{t} = mgv$$

$$E_p = (mgv)t$$